

FFGFT in Brief

Foundations, results and status after the calculation review

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FFGFT in Brief

Abstract

The Fundamental Fractal Geometric Field Theory (FFGFT) describes particles, couplings and gravitation as properties of a compact four-dimensional carrier $T^4/\mathbb{Z}_3$ whose fractal structure is fixed by exactly one dimensionless parameter, $\xi = 4/30000$. Its starting point is the time-mass duality $\tilde{T} \cdot m = 1$. This document summarises the entire corpus (Docs. 001–383, A series) as of 1 October 2026, that is, after the complete recalculation from 29 September to 1 October 2026. For every result it gives the formula, the numerical value with its comparison value, the status and the documents that carry it, and it states just as clearly what is set by choice, what remains open and what no longer holds. The core results: the lepton mass ratios follow from ξ to within 0.5 to 1.6 %; lepton masses, Planck energy, G and the minimal length L_0 hang on a single chain with exactly one measured value as anchor; charge quantisation follows algebraically from the Galois structure of $\text{GF}(27)^*$; the neutrino mass differences agree to 1 %; the Bell violation is preserved, and the ξ effect cannot be resolved with present-day hardware. The cosmic sector (H_0 , Λ) is deliberately set aside.

Status markers

- [B]** Algebraically proved (transformation or proof without additional assumption).
- [K]** Numerically confirmed (verification script, comparison with measurements).
- [S]** Choice, open or speculative: motivated but not derived.
- [X]** Refuted or excluded.
- [Q]** Convention or adopted quantity.

The corrected state is authoritative: the dated notes in the documents ("Corrected on", "Note of", "Removed on") and the correction register Doc. 190 up to R146. Comparison values: CODATA 2022, PDG 2024. All numbers in this document are recomputed by the verification script `pruef_384_zusammenfassung.py`.

1 What FFGFT is

Architecture in four layers

FFGFT rests on three explicitly chosen basic assumptions (R56, Doc. 365): the time-mass duality $\tilde{T} \cdot m = 1$, the compact carrier T^4 and the identification with order 3, $\mathbb{Z}_3$. A fourth, motivated structural choice is the lattice D_4 ; Doc. 380 shows that among the comparison carriers tested with a $\mathbb{Z}_3$ action that has fixed points, D_4 has the lowest lattice energy **[K]**, but the choice remains a choice **[S]**. In addition there is exactly one dimensionless parameter, $\xi = 4/30000$. The fine-structure constant is set to $\alpha = 1$ by adjusting the charge unit e ; this is a convention, not a result **[Q]**. Galois fields, Hodge theory and representation theory serve as consistency checks, not as additional inputs (Doc. 365, layer 4). The spacetime dimension $3 + 1$ is an empirical input (P34).

Guidelines

Three rules determine how results are to be read. First, FFGFT works with ξ alone; measured values from SI and the Standard Model serve only for conversion into units and for comparison. Exactly one measured value is needed as anchor: Doc. 383 shows that lepton masses, Planck energy, G , ℓ_P and L_0 hang on a single chain. Second, integer structural choices such as $\mathbb{Z}_3$, D_4 , the winding numbers or the exponent 10 in the cosmic sector are motivated choices, not calibrations. Third, the cosmic sector (H_0 , Λ) is deliberately excluded from the derivation of constants (P39): the Standard Model does not derive it either, and none of the frameworks compared determines the ratio of Planck length to Hubble length from first principles.

What the corpus no longer claims

The calculation review has withdrawn several earlier claims. FFGFT is not "postulate-free": $\tilde{T} \cdot m = 1$, T^4 , $\mathbb{Z}_3$, D_4 and ξ are chosen. Precision tables of older documents in which prefactors had been fitted to measured values are not evidence (R138). The agreement of α is a property of the chosen anchor, not an independent prediction (R135). The precision of G to 0.003 % is not evidence in its own right (R141). A ξ effect in Bell experiments or quantum gates that is measurable with present-day hardware does not exist (R142, R143). Where such statements appear in older documents, they carry a dated note.

2 Foundations

Time-mass duality

In natural units ($\hbar = c = 1$) the intrinsic time scale and the mass of a state are reciprocal:

$$\tilde{T} \cdot m = 1, \quad \tilde{T} = \frac{\hbar}{mc^2} \text{ (SI)}. \quad (1)$$

The relation describes every state as a snapshot, not as a law of motion (Doc. 365). A position-dependent mass in a gravitational potential is equivalent to a position-dependent time; "mass constant, space curved" and "space flat, mass variable" are two representations of the same situation (Doc. 078 **[K]**). On the carrier the duality holds mode by mode exactly as a position relation $\lambda_4 \cdot m = 2\pi$ (Doc. 314 **[K]**). Reciprocity gives $E \rightarrow \infty \Leftrightarrow T \rightarrow 0$; since time is bounded from below by $T_0 = \xi t_P$, the horizon is an extremum and not a singularity (Doc. 306).

The parameter ξ

$$\xi = \frac{4}{30000} = \frac{1}{7500} = 1.3 \times 10^{-4}. \quad (2)$$

ξ is the only dimensionless parameter. It has several consistent readings, but none produces the number without an input: geometrically as the number 4 of lattice directions of D_4 over $3 \cdot 10^4$ (Docs. 004, 320); as a Galois number via

$$\frac{(r_\mu/r_e)^2}{\xi} = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \quad (3)$$

(Docs. 336, 338), which is arithmetically exact **[B]** but depends on the chosen prefactors r_i ; and as the smallest eigenvalue of the D_4 sub-operator, $\xi = \lambda_{\min}(\hat{F}_{D_4})$ (Docs. 322, 327), which is well defined provided the D_4 sector lies in the $\mathbb{Z}_3$ -invariant sector **[B]**, but whose value is inserted, not derived. Overall, ξ is a motivated choice **[S]**: in the words of Doc. 365, not completely free, but not yet completely derived either. The Higgs geometry serves as a check: with the measured self-coupling $\lambda_h = m_h^2/(2v^2) = 0.129$ one obtains $\xi_{\text{EFT}} = \lambda_h^2 v^2 / (16\pi^3 m_h^2) = m_h^2 / (64\pi^3 v^2) = 1.30 \times 10^{-4}$, 2.6 % below $4/30000$ (Doc. 354, A190 **[K]**); the deviation is interpreted structurally **[S]** (R59). This is a consistency check to within a few per cent, not an exact derivation. Only the shortcut with $\lambda_h = m_h/v$ to $1/(16\pi^3)$, 15 times ξ (Doc. 320), does not hold **[X]**.

Fractal correction

$$K_{\text{frak}} = 1 - 100\xi = \frac{74}{75} = 0.98\bar{6}. \quad (4)$$

The value is confirmed **[K]** (R135): comparing the two routes to E_0 with the measured α requires $K = \xi m_e m_\mu \alpha^{-1} = 0.98650$, i.e. $n = 101.2$ instead of 100 in $1 - n\xi$. The residual of 1.7×10^{-4} is not explained by measurement uncertainties and is open **[S]**. The *form* of the factor (additive $1 - 100\xi$ or multiplicative $(1 - \xi)^{100}$) is open again **[S]** (R139): the available witnesses do not separate the two forms more sharply than their difference of 8.8×10^{-5} . The number 100 itself is chosen, not derived (note in Doc. 133, A040). K_{frak} acts only on absolute values and cancels exactly in ratios **[B]** (A040, Doc. 070). Equivalently, K_{frak} can be written in power form $(D/3)^{D/2} = 74/75$ with $D_f^{\text{eff}} = 2.97303$ **[K]** (Doc. 381); the spatial box dimension $D_f = 3 - \xi = 2.999867$ is a choice **[S]** that does not follow from the 100-step recursion (note in Doc. 322).

E_0 and the fine-structure constant

The characteristic energy is the geometric mean of the two light lepton masses, bare and with the fractal correction (R72):

$$E_0^{\text{bare}} = \sqrt{m_e m_\mu} = 7.348 \text{ MeV}, \quad E_0 = \sqrt{\frac{m_e m_\mu}{K_{\text{frak}}}} = 7.397 \text{ MeV}. \quad (5)$$

With $\alpha = \xi E_0^2$ (in MeV) one obtains $\alpha^{-1} = 137.06$, 1.7×10^{-4} away from the measured value. The Galois form $\alpha^{-1} = 3700/27 = 137.037$ (+7.6 ppm, Doc. 338, R102) follows from $(E_0^{\text{bare}})^2 = m_e m_\mu \approx 54 \text{ MeV}^2 = |\text{GF}(3)^*| \cdot |\text{GF}(27)| \text{ MeV}^2$ together with $K_{\text{frak}} = 74/75$. Both agreements are properties of the anchor E_0 , not independent predictions **[K]** (R135). Third routes to E_0 , closed α formulas from ξ alone and $E_0 = 1/\xi = 7500 \text{ GeV}$ are withdrawn **[X]**.

One chain, one anchor

Doc. 149 links the electroweak scale to the Planck energy; Doc. 383 inserts this into the lepton ladder:

$$\frac{\nu}{E_P} = \frac{\xi^4}{5\pi} = 2.012 \times 10^{-17} \text{ } (-0.23 \%), \quad m_i = \frac{r_i}{5\pi} \xi^{p_i+4} E_P \text{ } \mathbf{[B]}. \quad (6)$$

Thus ν drops out as an intermediate quantity, and a single measured value fixes all numbers. With m_e as anchor one obtains $E_P = 1.237 \times 10^{19} \text{ GeV}$ (+1.34 %), $G = 6.50 \times 10^{-11}$ (−2.6 %), $\ell_P = 1.595 \times 10^{-35} \text{ m}$ and $L_0 = \xi \ell_P$ **[K]**. The spread depending on the choice of anchor is the known residual of the ladder, not a second anchor:

Anchor	E_P	G	ℓ_P, L_0
m_e	+1.34 %	−2.62 %	−1.32 %
m_μ	+0.81 %	−1.60 %	−0.80 %
m_τ	−0.22 %	+0.45 %	+0.22 %
ν	+0.23 %	−0.46 %	−0.23 %

With the measured E_P the lepton masses themselves come out as 0.504, 104.8 and 1780.9 MeV (−1.32, −0.80, +0.22 %) **[K]**. Open is the factor 10 in $\nu = E_P/(f^4 \cdot \frac{\pi}{2} \cdot 10)$ with $f = 1/\xi$; it is motivated but not derived **[S]**.

Minimal length and minimal time

$$L_0 = \xi \ell_P = 2.155 \times 10^{-39} \text{ m}, \quad T_0 = \xi t_P = 7.19 \times 10^{-48} \text{ s}, \quad E_{\text{max}} = E_P/\xi = 9.16 \times 10^{22} \text{ GeV}. \quad (7)$$

The exponent 1 in $L_0 = \xi \ell_P$ is derived **[B]** (R90): L_0 is the Compton range of the time-field mediator with mass $m_T = M_{\text{basis}}/\xi$. The choice $M_{\text{basis}} = m_P$ in the fundamental sector is open **[S]**. L_0 makes the number of levels of the fractal operator finite and hence $\hat{F}$ bounded **[B]** (Doc. 327).

Time structure and recursion

The fractal correction runs step by step:

$$\xi_{n+1} = \xi_n (1 - 100 \xi_n), \quad \xi_0 = \frac{1}{7500}, \quad \xi_n \approx \frac{1}{100(n+75)}, \quad D(k) = \sum_{n < k} 100 \xi_n \approx \ln\left(1 + \frac{k}{75}\right). \quad (8)$$

In the frozen case with constant defect $d = 1/75$ the winding closes after exactly 75 turns; the rotation number is $74/75 = K_{\text{frak}}$ **[B]**. In the running case the time winding becomes a

logarithmic spiral with self-similarity ratio e , a continuous scale invariance with a pole at $n = -75$ (Doc. 295 **[B]/[K]**). The increments $d_n = 100 \xi_n$ decrease monotonically from $d_1 = 1/75$; d_1 is the largest, not the smallest increment, and the lower time bound is $T_0 = \xi t_p$ (Docs. 306, 307). Doc. 381 shows the telescoping product $\prod (1 - 100\xi_k) = \xi_n/\xi_0$ exactly **[B]**, a second order $1/\xi_n \approx 100(n + 75) + 100 \ln((n + 75)/75)$ that is 50 to about 1000 times more accurate **[K]**, and the constant $\psi'(75) = 0.01342$ between mass and time defect **[K]**. The assignment to a one-loop β function depends on the step convention **[S]**.

Geometry, Hilbert space, spectral theory

The state space is $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$; the translation between T^4 and $\mathcal{H}$ is bijective and lossless (Docs. 230, 270, 330). The torus is flat: the curvature vanishes pointwise via the metric descent **[B]** (R95). The spectrum separates the lattices: $\mathbb{Z}^4$ begins at $|k|^2 = 1$ with degeneracy 8, D_4 at $|k|^2 = 2$ with 24 (Doc. 314 **[K]**).

The fundamental fractal operator $\hat{F} = \bigoplus_n S_n \circ L_n$ with $S_n = r_n U_n$ and $r_n \in (0, 1)$ is bounded **[B]**. The $\mathbb{Z}_3$ pairing makes it normal; it is self-adjoint on the $\mathbb{Z}_3$ -invariant sector P_0 **[B]**, with deficiency indices $(0, 0)$ and exactly one realisation. On the full space only the real part $(\hat{F} + \hat{F}^\dagger)/2$ is self-adjoint; using it is a choice **[S]** (R145, Doc. 327). The particle masses are eigenvalues of the mass operator $\hat{M}$, not of $\hat{F}$ (Docs. 322, 330).

Galois structure

The finite fields $\text{GF}(3^n)$ are the algebraic language of the mode classification on $T^4/\mathbb{Z}_3$. One has $|\text{GF}(9)^*| = 8$, $|\text{GF}(27)^*| = 26 = 2 \cdot 13$, $|\text{GF}(81)^*| = 80 = 2^4 \cdot 5$; $\text{GF}(9)$ is not a subfield of $\text{GF}(27)$, both lie in the compositum $\text{GF}(729)$. The Frobenius $x \mapsto x^3$ splits $\text{GF}(27)^* \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ into two fixed points $\{+1, -1\}$ and eight three-element orbits; on $\mathbb{Z}_{13}$ four orbits arise:

$$O_1 = \{1, 3, 9\}, \quad O_2 = \{2, 5, 6\}, \quad O_3 = \{4, 10, 12\}, \quad O_4 = \{7, 8, 11\} \quad \textbf{[B]}. \quad (9)$$

The 5 is the only primitive prime divisor of $3^4 - 1$, not the smallest one (Docs. 336, 338, 341). With the IPI correspondent D. Matzke only the algebra is jointly confirmed (orders in $G(3)$ and $G(6)$, Doc. 341); the physical assignments are on the FFGFT side.

3 Particles

Lepton ladder

The charged lepton masses follow the ladder

$$m_i = r_i \xi^{p_i} \nu, \quad r = \left(\frac{4}{3}, \frac{16}{5}, \frac{25}{9}\right), \quad p = \left(\frac{3}{2}, 1, \frac{2}{3}\right), \quad p_g = \frac{3}{2} \left(\frac{2}{3}\right)^{g-1}, \quad (10)$$

with $r_g = n_{\phi,g}^2/n_{\theta,g}$ from the winding numbers $(3, 2)$, $(5, 4)$, $(9, 5)$ (Docs. 006, 317, 338) and a geometric sequence of exponents with starting value $D/2$ and ratio $2/3$ (Docs. 158, 018). The ratios do not contain ν :

$$\frac{m_\mu}{m_e} = \frac{12}{5} \xi^{-1/2} = 120\sqrt{3} = 207.85 \text{ (+0.52 \%)}, \quad \frac{m_\tau}{m_\mu} = \frac{125}{144} \xi^{-1/3} = 16.99 \text{ (+1.03 \%)} \quad \textbf{[K]}. \quad (11)$$

With $\nu = 246 \text{ GeV}$ the absolute values lie at -1.18 , -0.66 and $+0.37 \%$. Formula and ratio calculation are **[B]**, the agreement **[K]**; the r_i , p_i and winding numbers are choices **[S]** (R138). The step $1/3$ holds only from μ to τ ; e lies on an orbit of its own (Doc. 322). The ladder is written without K_{frak} (A100).

Galois identities, Koide and the four routes

$(m_\mu/m_e)^2_{\text{ladder}} = 43200$ is arithmetically exact; against the measured value a residual of -1.03% remains, an empirical discrepancy **[K]** whose cause is open **[S]** (R113). The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (12)$$

holds for the measured masses to $Q - \frac{2}{3} = -0.017\xi$ (PDG 2024; with $m_\tau = 1776.86$ MeV -0.046ξ , Doc. 352) **[K]**; as a prediction of m_τ from m_e, m_μ it lies 0.4σ from PDG, undecided. From the ladder, Koide follows only approximately ($+8\xi$). In the $\mathbb{Z}_3$ circulant form $\sqrt{m_k} = c + d\cos(\theta + 2\pi k/3)$ one has $Q = \frac{2}{3} \Leftrightarrow d/c = \sqrt{2}$ **[B]**. The matrix-element route with $|A| = \sqrt{2}/3$ and $\theta = |A|^2 = 2/9$ (Doc. 367) hits $m_\tau/m_e = 3477.47$ to 3×10^{-5} **[K]** but gives $m_\mu/m_e = 206.7703$, which is 442σ from CODATA; the exact pair $(\sqrt{2}, 2/9)$ is excluded for m_μ/m_e **[X]** (Doc. 369, R121). The geometric derivation of $d/c = \sqrt{2}$ in Doc. 353 does not hold **[S]**. Doc. 369 compares four routes to the lepton masses (ladder, Koide matrix element, Galois, φ skeleton); routes 1 and 3 are the same calculation in two languages for m_μ/m_e .

Electroweak scale

In the mass calculation v is an intermediate quantity. Depending on the route one obtains: 248.3 GeV with the Galois m_e from Doc. 338 (R104), 248.9 GeV with the measured m_e , 245.6 GeV with the measured m_e and K_{frak} , 245.65 GeV from $\xi^4 E_P/(5\pi)$ **[K]**. The comparison value $v = 246.22$ GeV is a Standard Model definition from G_F **[Q]** (R112); a convention-free comparison would only be possible via the muon lifetime τ_μ , which is open **[S]**.

Quarks

The quarks follow the same ladder form $m_q = r_q \xi^{p_q} v$ with topologically motivated exponents **[K]**, except for c and t , whose exponents are choices **[S]** (R138); the prefactors r_q are determined from measured values, their derivation is open **[S]** (R129, Doc. 373). The reproduction is therefore an encoding, not evidence:

Quark	r_q	p_q	Ladder ($v = 246$ GeV)	PDG 2024
u	6	$3/2$	2.27 MeV	2.16 MeV
d	$25/2$	$3/2$	4.73 MeV	4.70 MeV
s	$26/9$	1	94.8 MeV	93.5 MeV
c	2	$2/3$	1.284 GeV	1.273 GeV
b	$3/2$	$1/2$	4.26 GeV	4.18 GeV
t	$1/28$	$-1/3$	172.0 GeV	172.6 GeV

Light quark masses are scheme dependent; $\tilde{T} \cdot m = 1$ refers to pole masses. Λ_{QCD} does not follow from the ξ ladder **[X]** (R122): $v\xi^{1/3} = 12.6$ GeV, not 200 MeV. The top mass $m_t = (\frac{11}{8})^2 M_Z = 172.40$ GeV (-0.10%) is a candidate **[K]**; the CKM parameter $\lambda = \xi^{1/6} = 0.2260$ lies 0.46% from the measured value **[K]**; the exponent $1/6$ is a motivated choice **[S]** (note in Doc. 336). The quark and hadron sector as a whole is an open bridge (Doc. 318, R76).

Neutrinos

Doc. 340 derives hierarchy and mixing from the four Frobenius orbits. With the assumed base mass $m_\nu = \frac{\xi^2}{2} m_e = 4.54$ meV **[S]** one obtains

$$m_1 = m_\nu, \quad m_2 = \sqrt{\frac{14}{3}} m_\nu = 9.81, \quad m_3 = 11 m_\nu = 49.96 \text{ meV}, \quad \sum m = 64.3 \text{ meV}, \quad (13)$$

$$\Delta m_{\text{atm}}^2 = 120 m_\nu^2 = 2.476 \times 10^{-3} \text{ eV}^2, \quad \Delta m_{\text{sol}}^2 = \frac{11}{3} m_\nu^2 = 7.565 \times 10^{-5} \text{ eV}^2, \quad (14)$$

i.e. -1.0% (against 2.500×10^{-3} , Doc. 340) and $+0.46\%$ **[K]**, in normal hierarchy. The reactor angle gives $\sin^2 \theta_{13} = (25\xi)^{2/3} = 0.0223$ ($+1.4\%$) **[K]**. According to R108, ν_1 and ν_2 lie algebraically in the self-inverse Majorana layer and ν_3 is Dirac; this assignment of ν_3 to the orbit O_4 is in tension with Doc. 346, which assigns O_4 to the charged leptons, and Doc. 344 lists all neutrinos as Dirac **[S]**; that the algebraic self-inversion physically means $\nu = \bar{\nu}$ is a further assumption **[S]**. For neutrinoless double beta decay this gives $m_{ee} \approx 6.0 \text{ meV}$ (phases zero), 0.13 meV in the destructive case, at most 14.4 meV , and exactly zero if all neutrinos are Dirac (Doc. 344). Older neutrino spectra (Docs. 006, 007) are excluded **[X]** (R109; note in Doc. 007). The A_5 eight-element spectrum gives $\sin^2 \theta_{23} \in \{4/9, 5/9\}$; $5/9 = 0.556$ lies close to the measured value for normal hierarchy, so maximal mixing $1/2$ is excluded (Doc. 372) **[K]**. θ_{12} is captured only in order of magnitude, the CP phase not at all **[S]**. The venting exponent $p = 5/24$ follows from the decomposition $\rho_4 \downarrow_{A_4} = \rho_3 \oplus \rho_1$ with $|A_5 : A_4| = 5$ **[B]** (Doc. 340).

Anomalous magnetic moments

The ratio line carries the result **[K]** (Docs. 018, 158): with winding exponents $5/3$ and $4/3$, $\Delta a(\tau-\mu)/\Delta a(\mu-e) = f^{1/3} - 1 = 18.57$, and anchored to the measured a_e, a_μ one obtains

$$a_\tau = a_e + \frac{144}{125} \frac{m_\tau}{m_\mu} (a_\mu - a_e) = 1.2811 \times 10^{-3}, \quad (15)$$

1.04×10^{-4} above the Standard Model value; present bounds are about 85 times wider, the decision is pending **[S]**. A fixed FFGFT addition $\Delta a_\mu = 251 \times 10^{-11}$ as the explanation of the former muon anomaly is superseded by the 2025 status; it would lie 3.4σ above today's difference **[X]** (Doc. 382).

Electroweak sector, Higgs, top

On shell, $\sin^2 \theta_W = 1 - M_W^2/M_Z^2 = 2/9$ holds as an element of the A_5 eight-element spectrum (Doc. 372) **[K]**; this gives $M_W = M_Z \sqrt{7/9} = 80.42 \text{ GeV}$, 3.8σ from the 2024 world average, an explicitly stated, falsifiable tension. In the $\overline{\text{MS}}$ scheme the renormalisation running from $3/8$ gives $\sin^2 \theta_W(M_Z) = 0.2308$ (-0.19%) with two external inputs (Doc. 323) **[K]**. The Higgs mass $m_h = \frac{11}{8} M_Z = 125.38 \text{ GeV}$ ($+0.15\%$) and the trace rule $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ ($+0.45\%$) are candidates **[K]**; the origin of the factors is a reading **[S]** (Doc. 373). The chain "symmetry breaking topologically enforced" is open at two points **[S]** (Doc. 355). Earlier Higgs formulas ($m_h = v\xi^{1/4}$, Doc. 041) are refuted **[X]** (R122).

Charge, hypercharge, generations, colour, photon

Charge quantisation follows algebraically **[B]** (Doc. 346, R111): the QR orbits $O_1, O_3 \bmod 13$ are electrically neutral (neutrinos, antineutrinos), the NQR orbits O_4 and O_2 are charged (charged leptons, up quarks); together with $N \bmod 3$ this gives $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$, and "neutral and coloured" is excluded. The down quarks have no orbit of their own. Gell-Mann–Nishijima is not fully closed: $I_3 = \pm\frac{1}{2}$ follows **[B]**, but the hypercharge of the charged leptons does not follow from the QR/NQR bit alone and is open **[S]** (R146, Doc. 347). The three elements of a Frobenius orbit give three generations **[B]** (Doc. 348); the earlier swap of generations 2 and 3 is an ordering artefact. $\mathfrak{su}(3)$ arises from the $\mathbb{Z}_3$ eigensectors **[B]**; $SU(3)$ is the smallest, not the only suitable group (Doc. 321). Photon and gluons are massless **[B]**; that masslessness means spin 1 does not follow from this and is open **[S]** (Doc. 339).

4 Gravitation and cosmos

Gravitational constant

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}, \quad C_{\text{conv}} = 7.783 \times 10^{-3}, \quad K_{\text{frak}} = 0.986. \quad (16)$$

G is not an independent constant: its form and order of magnitude follow from ξ and the one anchor **[K]** (Docs. 012, 180, 383). C_{conv} is the SI conversion of the one-anchor chain; its corpus value additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The agreement with CODATA to 0.003 % arises only with this residual and is not precision evidence in its own right (R141); it holds with the rounded corpus value $K_{\text{frak}} = 0.986$ of the G formula, with 74/75 it would be +0.07 %. Explaining the residual structurally is open **[S]**. From $\tilde{T} \cdot m = 1$ one has $m \cdot G = \xi^2/4$; a mass singularity with finite G is excluded **[B]** (Doc. 350), while physical validity in the black-hole regime is open **[S]**.

Black holes

Equating the Compton and Schwarzschild radii gives, purely geometrically,

$$M_{\text{coll}} = \frac{m_P}{\sqrt{2}}, \quad r_s = \sqrt{2} \ell_P, \quad S_{\text{BH}}(M_{\text{coll}}) = 2\pi \text{ nat} = \frac{2\pi}{\ln 2} \text{ bit} = 9.065 \text{ bit} \quad \mathbf{[B]} \quad (17)$$

(Docs. 325, 329). S_{BH} equals Matzke's length in Planck units; Matzke's threshold is smaller by $\sqrt{2}$, $n_{\text{thr}} = 6.41$ (R144). As a universal bit number n_{thr} is refuted **[X]** (R88). The ratio of sector to thermal information is mass independent, $\ln 3 \approx 1.099$ **[B]**. The Hawking temperature corresponds to the bit energy $\hbar c/L$ at the horizon scale **[K]**; only modes with triality 0 can be emitted, and confinement and Hawking selection are one principle **[B]** (Doc. 325). Greybody factors and back-reaction are open.

Casimir and CMB

The Casimir energy density between ideal plates is $-\pi^2 \hbar c / (720 d^4)$; the form with 240 is the pressure. At the length L_ξ the ratio to the CMB density, $\pi^2 / (720 \xi) = 102.8$, is an identity, because L_ξ itself is determined from the CMB density and ξ ; it is not measurement evidence (R136). The QFT Casimir energy does not supply a factor 4/3 **[X]** (Doc. 371). The temperature $k_B T_{\text{CMB}} = \frac{16}{9} \xi (1 - \frac{275}{4} \xi) \text{ eV}$ matches only in the unit eV and with an underived coefficient **[S]** (R137). The factor 3 in front of the peak sequence $\{1, 6, 14, 26\}$ is a choice; the sequence itself is read off the data (R139).

Cosmic sector: deliberately set aside

The structural form $H_0/v_e = \pi^2 \xi^{10} = 1.75 \times 10^{-38}$ gives $H_0 = 66.8 \text{ km/s/Mpc}$; the exponent 10 is a motivated choice (P20), so the statement is conditional and not a prediction. Equally conditional are $\Lambda^* = 1/R_H^2$ and the recasting of the 10^{123} question as a structural question (Doc. 312). Excluded are the Λ formulas of older documents, $H_0 = \xi c$, $z = \xi x$ and $z \approx \xi \ln(d/\ell_P)$ **[X]** (R137; note in Doc. 156). The redshift $1+z = e^{\xi x}$ is achromatic **[K]**, but the length in x is not specified from ξ ; the supernova time dilation is $1+z$ for all z **[B]** (R128). Supernovae, BAO and the CMB temperature are degenerate between expansion and intrinsic time unfolding; the decision is theoretical, not empirical (Doc. 267).

Galaxies

The acceleration scale $a_0 = c^2 \xi^{10} / (4\bar{\lambda}_e) = 1.03 \times 10^{-10} \text{ m/s}^2$ lies 14 % below the empirical value and inherits the choice P20. The gas-dominated dwarf galaxy DDO 154 matches to 0.2 % **[K]**; for spiral galaxies deviations of about 19 % remain open. “Bullet Cluster passed” is not supported: with baryonic masses the peak lies at the gas (note in Doc. 308). Light deflection reproduces the GR value 1.75” **[K]**.

5 Quantum mechanics

FFGFT reads a measurement as a deterministic pole transition in state space, $A(z, \lambda) = \text{sgn}(z - \lambda)$ with $P(+1) = (1 + z)/2$; randomness is epistemic. For several qubits every such reading is non-local or contextual by Bell and Kochen–Specker, and the T^4 reading satisfies this: the correlation is a property of the closed geometry (Doc. 230, R126). Operationally, FFGFT and standard quantum mechanics give the same predictions up to a ξ correction.

The Bell violation remains: the fractal damping reduces S only by an amount of order ξ , $\Delta S/S \approx 10^{-5}$ to 10^{-4} depending on the form, and establishes no local bound (R142). The only documented hardware measurement (IBM, 28 May 2026, Doc. 147, 2048 shots per run):

Device	Runs	$\langle S \rangle$	Standard error	Share of $2\sqrt{2}$
ibm_kingston	15	2.7396	0.0071	96.9 %
ibm_marrakesh	10	2.7016	0.0138	95.5 %
pooled	25	2.7244	0.0078	96.3 %

The Bell violation is confirmed **[K]**; the gap to $2\sqrt{2}$ (0.088 on ibm_kingston, 0.104 pooled) is decoherence, 160 to several thousand times larger than the ξ effect, and two identical chips differ by 0.038. The ξ effect cannot be resolved with present-day hardware **[S]**. CHSH values given earlier from alleged runs in 2025 and a ξ value fitted to them are unsupported and removed (R143). The deviation of the π gate by $100\xi = 1.33\%$ is absorbed by the pulse calibration of real qubits and is not directly measurable (R142). A factorisation method distinguishable from Shor’s does not exist **[X]**; ξ as a line width of phase estimation is refuted for the preregistered setup, the line remains open (P44).

6 Testable statements

Quantity	FFGFT	Measured / bound	Status
m_μ/m_e	207.85	206.77 (+0.52 %)	[K] , r_i [S]
m_τ/m_μ	16.99	16.82 (+1.03 %)	[K]
Koide Q	2/3	-0.017ξ	[K]
m_τ from Koide	1776.97 MeV	1776.93(9), 0.4σ	open
v/E_P	$\xi^4/(5\pi)$	-0.23%	[K] , factor 10 [S]
Δm_{atm}^2	$2.476 \times 10^{-3} \text{ eV}^2$	$2.500 \times 10^{-3} (-1.0\%)$	[K]
Δm_{sol}^2	$7.565 \times 10^{-5} \text{ eV}^2$	+0.46 %	[K]
$\sin^2 \theta_{13}$	0.0223	0.0220 (+1.4 %)	[K]
$m_{ee} (0\nu\beta\beta)$	6.0 / 0.13 / 0 meV	nEXO 5–20 meV	[S] , nEXO decides
$\sum m_\nu$	64.3 meV	cosmological, model dependent	[K]

Quantity	FFGFT	Measured / bound	Status
a_τ	1.2811×10^{-3}	bounds 85 times wider	open
M_W from 2/9	80.42 GeV	3.8σ (world average)	tension
m_h	125.38 GeV	125.20 (+0.15 %)	candidate [K]
m_t	172.40 GeV	172.57 (−0.10 %)	candidate [K]
λ_{CKM}	$\xi^{1/6} = 0.2260$	0.2250 (+0.46 %)	[K], exponent [S]
τ_μ without G_{Ft} ν	–	MuLan	open [S]
CHSH deviation	$\Delta S/S \sim 10^{-5} - 10^{-4}$	hardware noise $\sim 10^{-2}$	not resolvable
π gate	1.33 %	in pulse calibration	not measurable

7 What no longer holds

The following statements of older documents are refuted or withdrawn [X]; the documents carry dated notes on them. In the foundations: ξ as necessarily derived, the shortcut of the Higgs route to $1/(16\pi^3)$ (the Higgs check to 2.6 % remains), a Casimir factor 4/3, the number 100 as derived, a minimal time increment d_1 , self-adjointness of $\hat{F}$ on the full space, a field tower $\text{GF}(3) \subset \text{GF}(9) \subset \text{GF}(27)$. For particles: precision tables with fitted prefactors as evidence, the φ mass ladder, the exact pair $(\sqrt{2}, 2/9)$ for m_μ/m_e , Λ_{QCD} from the ξ ladder, earlier Higgs formulas, the neutrino spectra of Docs. 006 and 007, a fixed muon addition $\Delta a_\mu = 251 \times 10^{-11}$, Gell-Mann–Nishijima as fully closed. In gravitation and cosmos: the precision of G as evidence in its own right, n_{thr} as a universal number, the Casimir-CMB ratio as measurement evidence, the Λ formulas, $H_0 = \xi c$ and $z = \xi x$, "Bullet Cluster passed". In quantum mechanics: local realism or $S \leq 2 + \varepsilon$, a hardware confirmation of the ξ effect, the CHSH values 2.8275 and 2.8278, a factorisation method specific to FFGFT.

8 Open bridges

Doc. 190 keeps the open points in a list of bridges (as of R146). In summary:

1. Foundations: the residual 1.7×10^{-4} between the K_{frak} required by α and $1 - 100\xi$, and the form of the factor (R135, R139); the residual of the one-anchor chain for G (R141) and the factor 10 in ν/E_p (Docs. 149, 383); $M_{\text{basis}} = m_p$ (R90); self-adjointness of $\hat{F}$ on the full space (R145); an adversarial D_4 specificity test (R95); the spectral dimension $d_s = 1.86$ versus ≈ 2 .
2. Particles: τ_μ directly from FFGFT as the only convention-free test of the weak sector (R112); the cause of the decreasing lepton residuals (R113, R121); the Koide amplitude $d/c = \sqrt{2}$ from the geometry; the hypercharge of the charged leptons (R146); the quark prefactors and the hadron sector (R76, R129); K_{had} (R123); two-loop running of α_s ; m_{Pl} and $\alpha_{\text{em}}(M_Z)$ from ξ ; spin 1 of the photon.
3. Gravitation and cosmos: greybody factors and back-reaction (R85); a physical Casimir signature of L_ξ (R136); T_{CMB} independent of units (R137); forward selection of the CMB peaks and the C_ℓ projection (P29–P31); the exponent 41/4 or 10 forward (P20); light elements without a hot early phase; Ω_{DM} model-neutrally.
4. Method: ξ as a line width (P44); clean-up of the H_0 language (P16, P34).

9 Status balance

Status	Results (selection)
[B]	Charge quantisation from the Legendre symbol (Doc. 346); $SU(2)_L$, $U(1)_Y$ and $\sin^2 \theta_W _{\text{GUT}} = 3/8$ (R80); $\mathfrak{su}(3)$ from $\mathbb{Z}_3$ sectors (Doc. 321); three generations as a Frobenius orbit (Doc. 348); self-adjointness of $\hat{F}$ on P_0 (R145); exponent in $L_0 = \xi \ell_P$ (R90); $m_i = \frac{\xi_i}{5\pi} \xi^{p_i+4} E_P$ (Doc. 383); $S_{\text{BH}}(M_{\text{coll}}) = 2\pi \text{nat}$ and $\ln 3$ (R144); supernova time dilation $1+z$ (R128); telescoping product of the recursion (Doc. 381).
[K]	Lepton ratios and Koide; neutrino mass differences and θ_{13} ; a_τ ratios; v/E_P ; value $K_{\text{frak}} = 74/75$; form and order of magnitude of G ; Higgs and top candidates; λ_{CKM} ; Higgs check of ξ to 2.6 %; Bell violation on hardware.
[S]	$\tilde{T} \cdot m = 1$, T^4 , $\mathbb{Z}_3$, D_4 , ξ_i , r_i , p_i and winding numbers; the number 100 and the form of $K_{\text{frak}i}$; factor 10 in v/E_P ; $M_{\text{basis}} = m_P$; base mass m_v ; quark prefactors; hypercharge of the charged leptons; spin 1 of the photon; exponent 10 in the cosmic sector; CMB factor 3.
[Q]	$\alpha = 1$ with adjusted e ; $v = 246 \text{ GeV}$ as Standard Model definition; the anchor m_μ/m_e is QED dependent; Maxwell dynamics adopted; C_{conv} as SI conversion.
[X]	see section "What no longer holds".

10 Method and bookkeeping

The corpus labels every statement with a status marker. Newer documents have a verification script with assertions, many of them null tests. The register Doc. 190 records corrections to older documents and open bridges; since R133 the source documents themselves are corrected, each passage with a dated note and the previous value. Excluded predictions are not rewritten but annotated. From 29 September to 1 October 2026 the entire corpus was recalculated: about 235 documents now carry a correction box, the A series carries notes in the text. The consequences for the classification are recorded in R135 to R146. Guard rails: a comparison value is not an input; a form $X = \xi^N$ alone says nothing (P35); predictions are preregistered with fixed degrees of freedom (P44); every prediction names its anchor (R113).

11 Reading guide

Topic	Documents
Foundation	365 (four layers), 190 (register), 180/189 (chain $\xi \rightarrow L_0$), 383 (one anchor)
Masses	006 (ladder), 336/338 (Galois), 351/352 (PDG, lepton residuals), 367/369 (matrix element, four routes)
Electroweak	323 (Weinberg running), 372/373 (2/9, Higgs, Yukawa), 355 (symmetry breaking)
Charge, colour	321 ($SU(3)$), 346/347 (charge, Gell-Mann–Nishijima), 348 (generations)
Neutrinos, $g - 2$	340, 343/344 (Majorana/Dirac), 158/382 (a_τ , status 2025)
Gravitation	012/013 (G , SI), 325/329 (black holes), 350
Structure	314 (lattices), 322/327/330 (spectral theory), 380 (D_4), 381 (recursion)
Time	295 (log spiral), 306/307 (reciprocity, time in state space)
Quantum mechanics	230/231 (Hilbert space, measurement), 023 (Bell), 147 (hardware), 175 (gates)
Delimitation	309 (scale anchor), 267 (degeneracy), 377 (interfaces), 378 (peer review)

12 Conclusion

FFGFT replaces the free parameters of the lepton sector by a structure with one parameter ξ and chosen integers, connects masses and the Planck scale through a chain with a single anchor, and obtains charge quantisation algebraically. The ratios agree to within a per cent down to per mille; the absolute values carry a known residual. Open are the origin of the number 100 and of the factor 10, the form of the fractal correction, the hypercharge of the charged leptons, the hadron sector and everything that depends on the cosmic sector. Decisive tests in the coming years are m_{ee} (nEXO), a_τ (Belle II), the sum of neutrino masses and M_W . The corpus labels choices as choices and refutations as refutations; this summary reflects that state.

Verification script: 2/python/Dok384_Skripte/pruef_384_zusammenfassung.py — all numbers in this document recomputed.